

TRUNG VU

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Education

Ph.D. in Mathematics, Yale University

2021– Present

Advisor: Prof. Ivan Losev

B.Sc. in Mathematics, HKUST

2017–2021

Research Interests

Geometric representation theory, Quantum Algebra, quantum groups at roots of unity, Soergel bimodules and related objects.

Publications

- T. Vu, *On the functor relating Harish-Chandra bimodules and Soergel bimodules*. arXiv: 2507.17067
- I. Losev, A. Tsymbaliuk, T. Vu, *On De Concini-Kac forms of quantum groups*. arXiv: 2601.06696
- T. Vu, *Quantum Harish-Chandra bimodules at roots of unity and affine Hecke category*. arXiv: 2606.26331

Fellowships & Awards

- **Conference Travel Fellowship**, Yale University *2026*

Invited Talks

- AMS Fall Central Sectional, University of Minnesota, Duluth *Oct 2026*
- Representation and Lie Theory seminar, University of California, Riverside *May 2026*
- AMS Eastern Sectional Meeting, Boston College, [slides](#) *Mar 2026*
Quantum Harish-Chandra bimodules at roots of unity
- Seattle Noncommutative Algebra Conference, University of Washington, [slides](#) *Dec 2025*
On De Concini-Kac forms of quantum groups
- Physics and Math seminar, Purdue University *Mar 2025*

Contributed Presentations

- Categorification and Symplectic Duality Workshop, Northeastern University (Poster) *Jun 2025*

Teaching

Instructor of Record, Yale University

- Math 1200 Calculus of Functions of Several Variables Spring 2027
- Math 1150 Calculus of Functions of One Variable II Fall 2025
- Math 115 Calculus of Functions of One Variable II Spring 2025

Teaching Assistant

- Math 115 Calculus of Functions of One Variable II Spring 2024
- Linear algebra Spring 2023
- Math 112 Calculus of Functions of One Variable I Spring 2022

Grader

- Calculus, Linear algebra, Abstract algebra, Introduction to differential manifold.

Mentoring

- SUMRY *Summer Undergraduate Math Research at Yale* Summer 2026

Workshops and Conferences Attended

- WARTHOG 2025, University of Oregon, *Cluster Algebras and Braided Varieties* Aug 2025
- 7-th Canada-Mexico-US meeting, University of Southern California, *Representation Theory, Noncommutative Algebra, and Categorification* Jul 2025
- Advances in Representation Theory, Northeastern University Jun 2025
- Quantum Groups and Representation Theory, North Carolina State University, *On the occasion of Kailash Misra's 70 birthday* Oct 2024
- WARTHOG 2024, University of Oregon, *Coherent-constructible equivalences in local Geometric Langlands and Representation Theory* Jul 2024
- MSRI summer school, *Koszul Duality in the Local Langlands Program* Jul 2024
- The 14th Southern Lie Theory Workshop series, University of Virginia, *Quantum structures on Lie Theory* Mar 2024
- Summer school in Geometric Representation Theory, MIT, *Coulomb Branches and Knot Homology* Jun 2023